

自动控制原理II

(Principle of Automatic Control Theory)

黄 剑: huang_jan@hust.edu.cn

学时: 40学时

考试: 闭卷

References

1. 王永骥，王金城，王敏主编。自动控制原理。北京：化学工业出版社， 2007
2. 胡寿松主编。自动控制原理。第三版。 北京：国防工业出版社， 1994
3. **Automatic Control Systems, Benjamin C.Kuo, Farid Golnarghi**,第8版，影印版，高等教育出版社,2003
4. **Modern Control Systems, Richard C.Dorf, Robert H.Bishop**,第9版，影印版，科学出版社， 2002

Chapter 7 Analysis and Design of Linear Discrete-Time System

7.1 Introduction

7.2 The Sampling Process and Sampling Theorem

7.3 Signal Recovery and Zero-Order Hold

7.4 Z-Transform and Inverse Z Transform

7.5 Mathematical Models of Discrete-Time Systems

7.6 Dynamic Performance of Discrete-Time Systems

7.7 Digital Control Design for Discrete-Time Systems

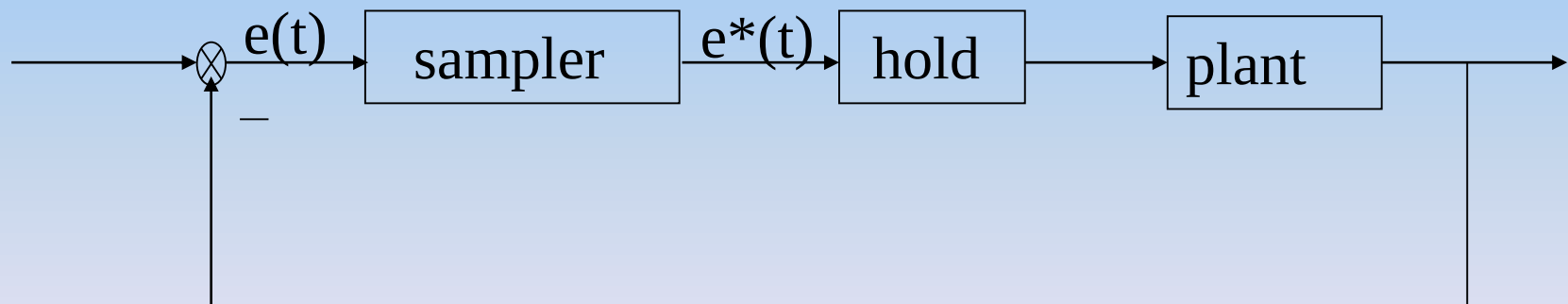
7.1 Introduction

Discrete-Time Systems:

Types: $\begin{cases} \text{Sampled data systems: Discrete Time, Continuous Value} \\ \text{Digital systems: Discrete Time, Quantized Value} \end{cases}$

Sampled Data System: a system that is continuous except for one or more *sampling operations*.

Digital System: There is one or more *digital signals* in the system.



Sampled-data control system

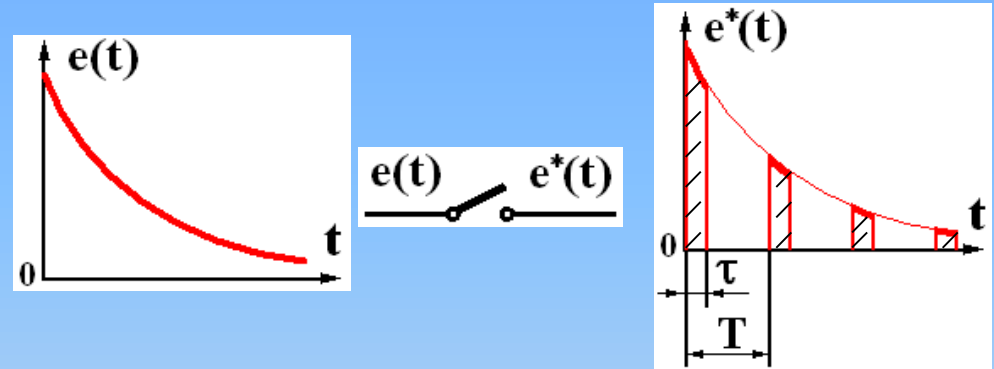
$e^*(t)$ is obtained by sampling a continuous signal $e(t)$.

A/D : analog to digital converter

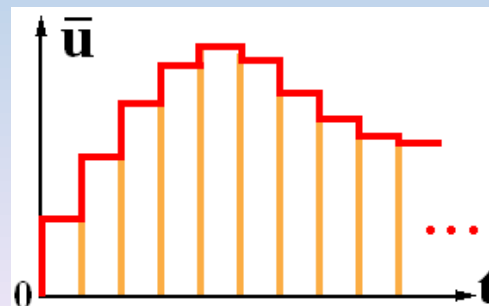
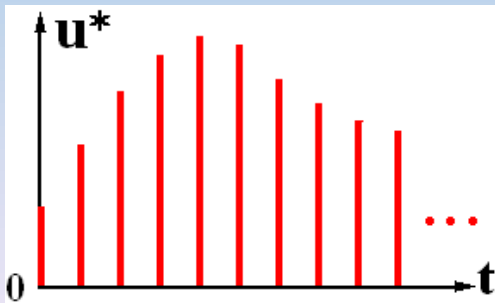
D/A : digital to analog converter

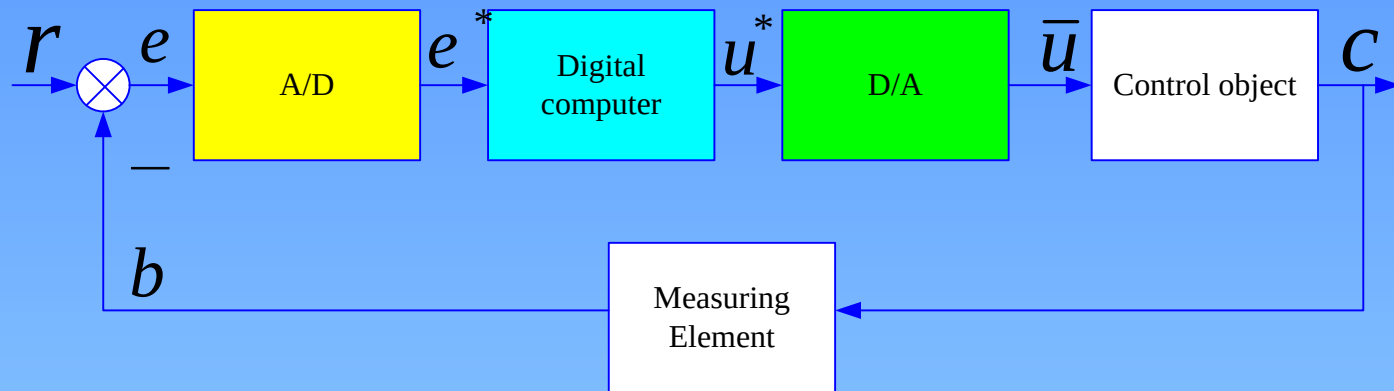
A/D process

- **Sampling** — Time sampled
- **Quantization** — Value quantized

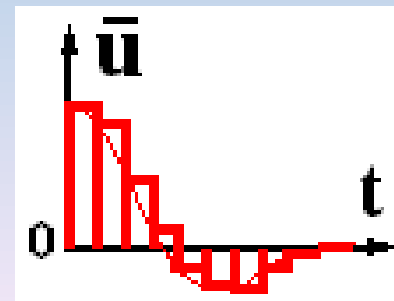
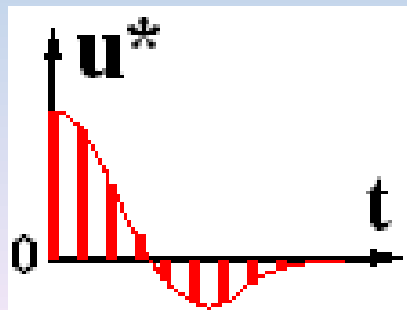
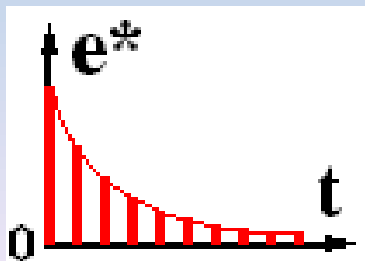
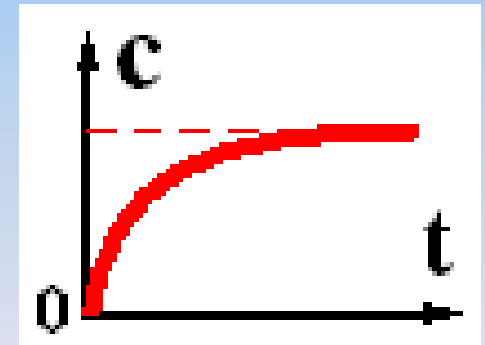
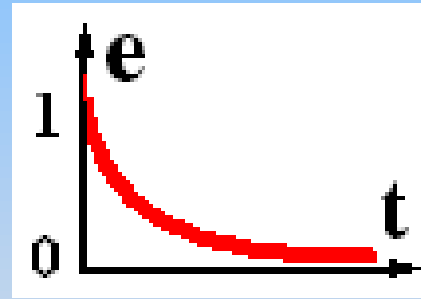
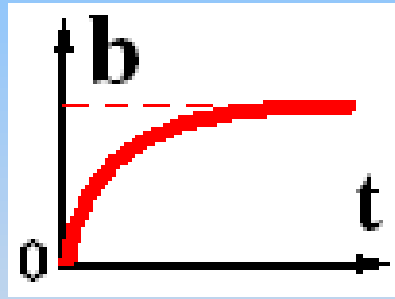
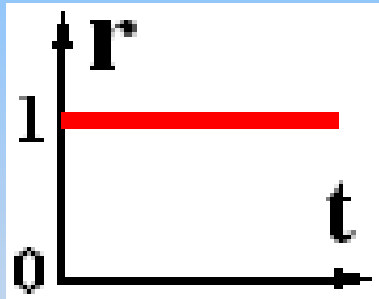


D/A process





Computer Controlled Systems



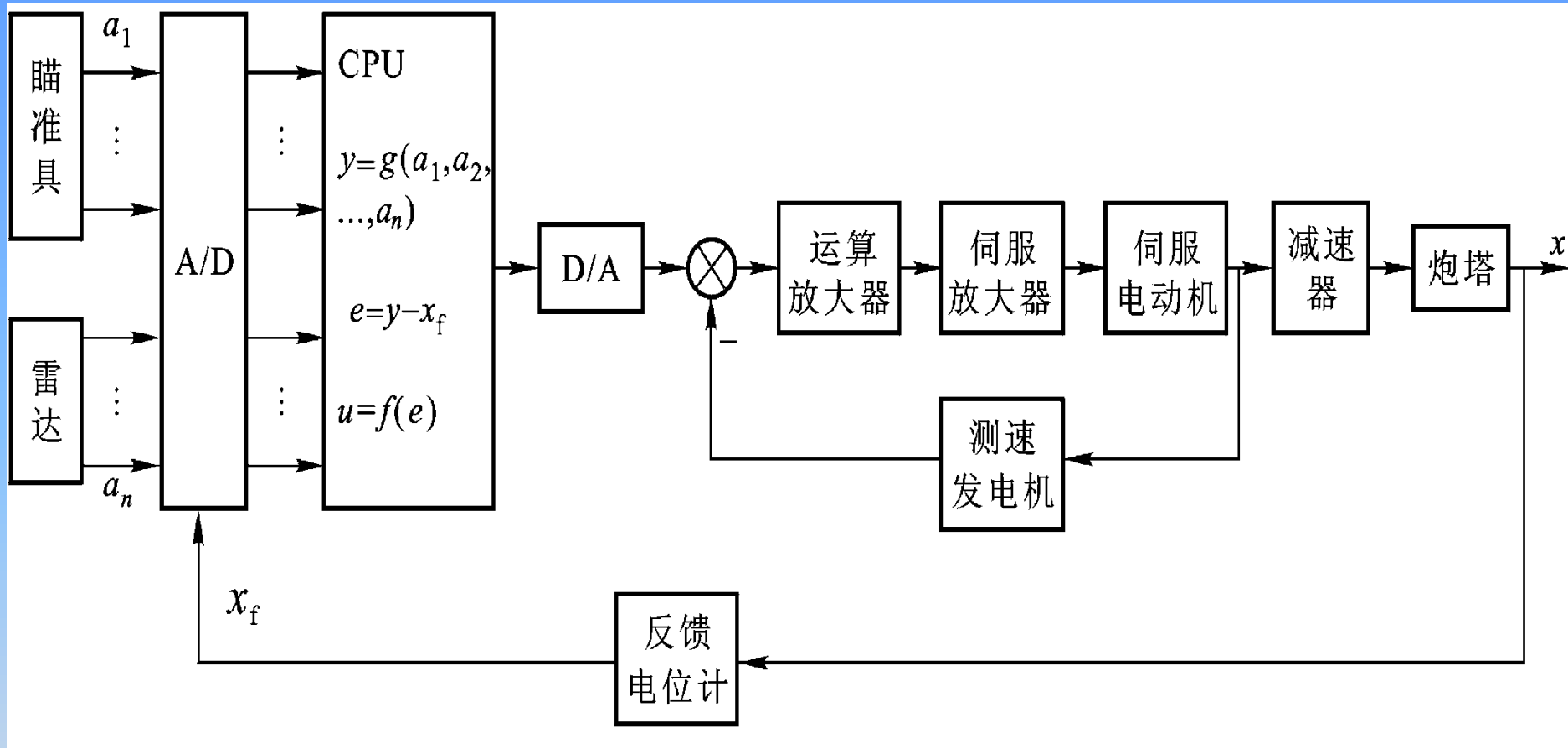


Fig 7-1 机载火力控制系统原理

History of Discrete-time system (p. 212-213)

DDC-Direct Digital Control （直接数字控制系统）

SCC- Surveillance Computer Control System(计算机监督控制系统)

TDC- Total and Distributed Control(集散控制系统):
muti-agent robots

Advantages and Disadvantages

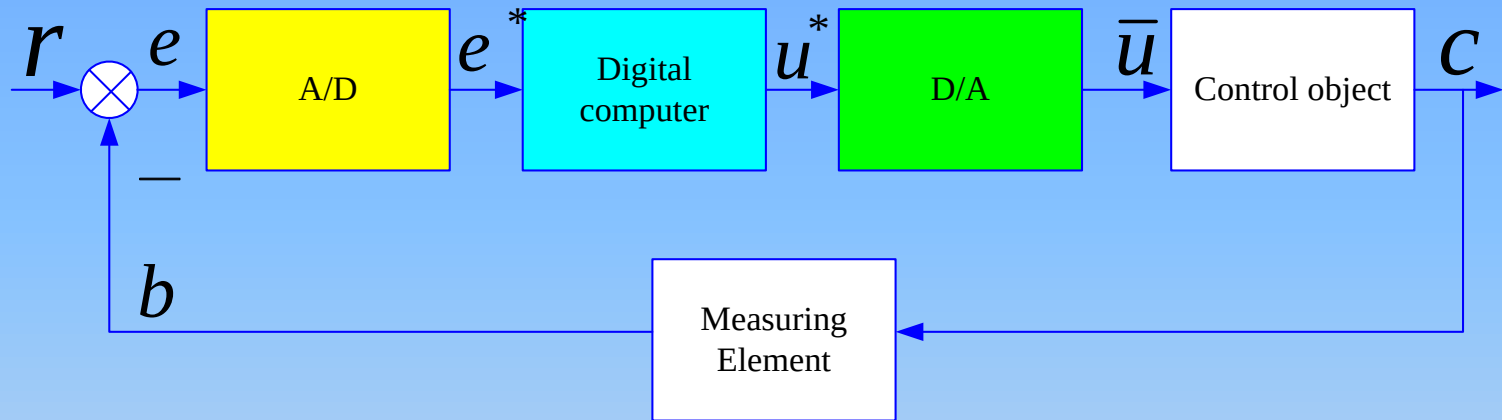
Computer Control System

- {
 - (1) Calculations are performed in the software. Easy for modification.
 - (2) Complex control laws easily realized;
 - (3) Reduced sensitivity to noise;
 - (4) One computer for multi-tasks, high utilization ;
 - (5) Network for process automation, macro-management and remote control.

- {
 - (1) Information between samples is lost. Compared with continuous system in the similar condition, the performance is reduced
 - (2) Needs A/D and D/A conversion devices

7.2 The Sampling Process and Sampling Theorem

7.2.1 The Sampling Process

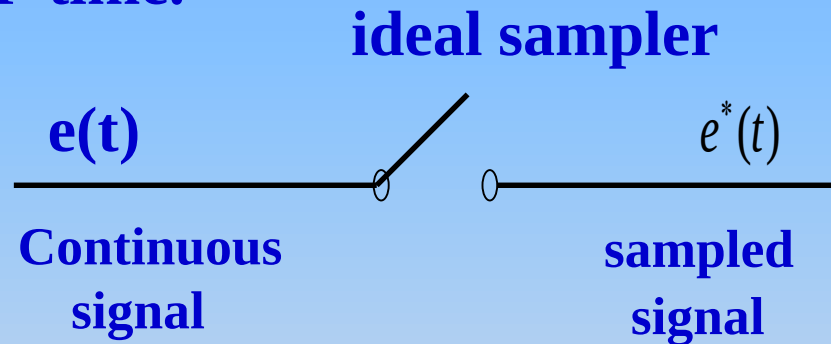


Computer Control System

Question: In the above computer control system, which signals are discrete, which signals are continuous?

- **Sampling Process:** Continuous signal $\rightarrow$ Discrete Signal
- **Holding Process:** Discrete Signal $\rightarrow$ Continuous Signal.
- The two are inverse process to each other.

Sampler: A switch that closes every T seconds for one instant of time.

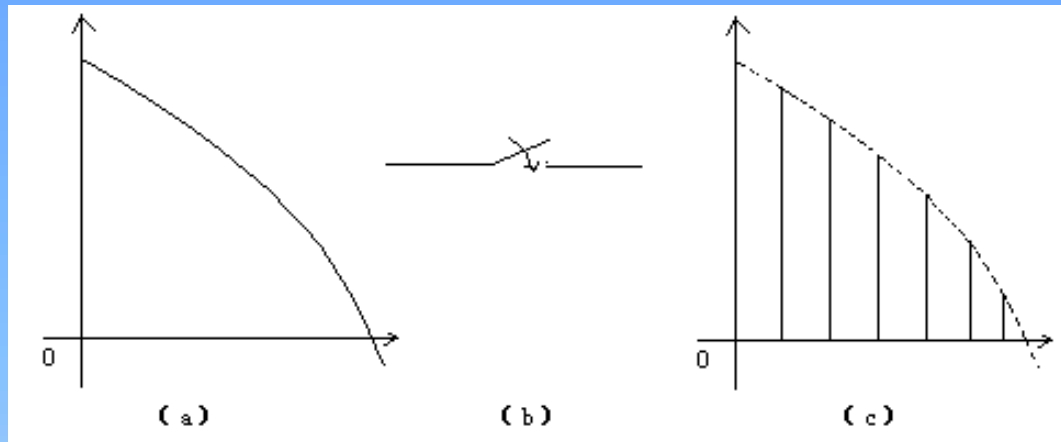


Where T is called the sampling period.

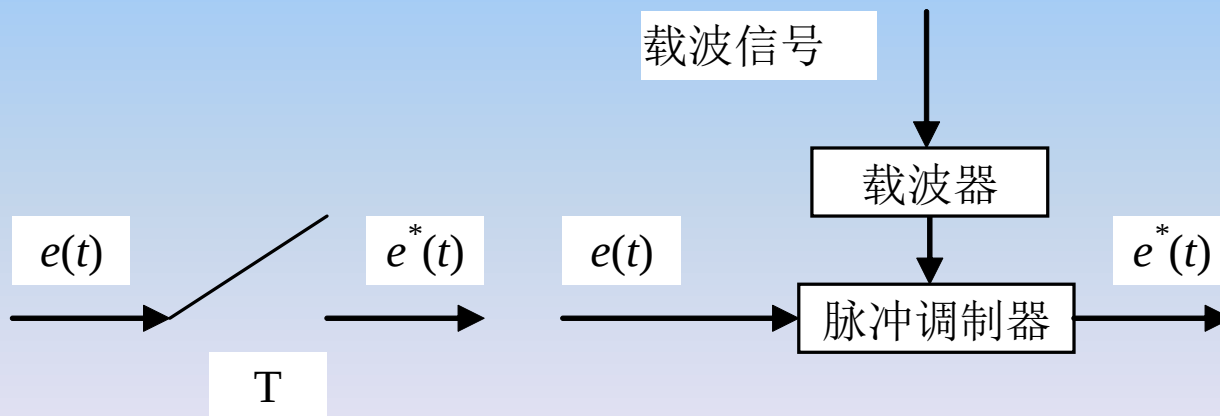
Ideal sampling process:

- (1) $t \ll T$. The sampling process is completed instantaneously
- (2) Word Length is enough, thus $e^*(Kt) = e(Kt)$

Types of Samplers: ideal, periodical, random,...



Sampling Process



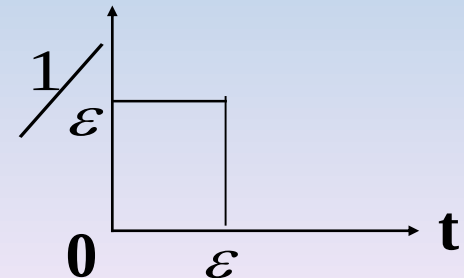
7.2.2 Mathematical Model for Sampling Signals

1、 Some ideal assumptions

- The sampler can be connected and cut down immediately;
- The signals in and out the sampler have no error/noise;
- $\tau \ll T$, that is $\tau \rightarrow 0$;
- The output is constant when sampler shuts down;
- Sample Period T is a constant.

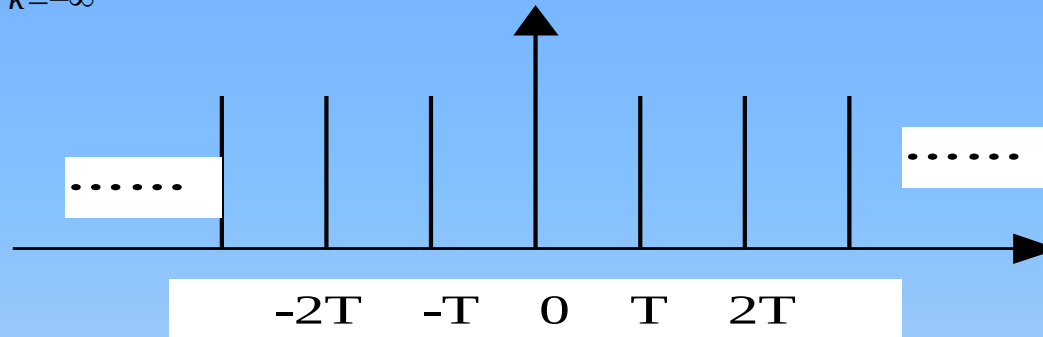
2、 Unit Impulsive Signal $\delta(t)$

$$\delta(t) = \begin{cases} \frac{1}{\varepsilon} & 0 \leq t \leq \varepsilon \\ 0 & t < 0 \text{ or } t > \varepsilon \end{cases}$$



3、Unit Impulsive Sequence Signal

$$\delta_T(t) = \sum_{k=-\infty}^{\infty} \delta(t - kT) = \delta(t) + \delta(t - T) + \cdots + \delta(t - kT) + \cdots$$



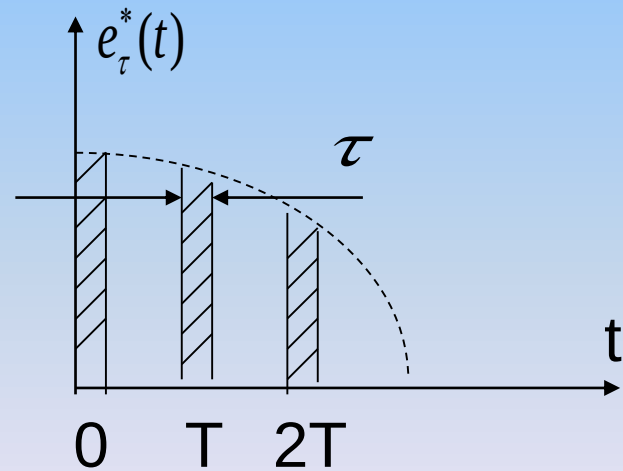
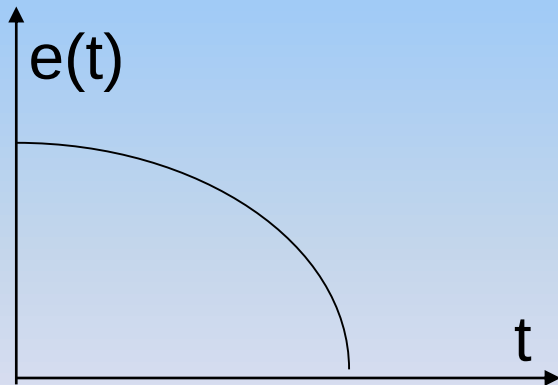
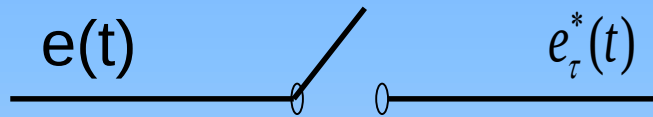
Unit Impulsive sequential signal

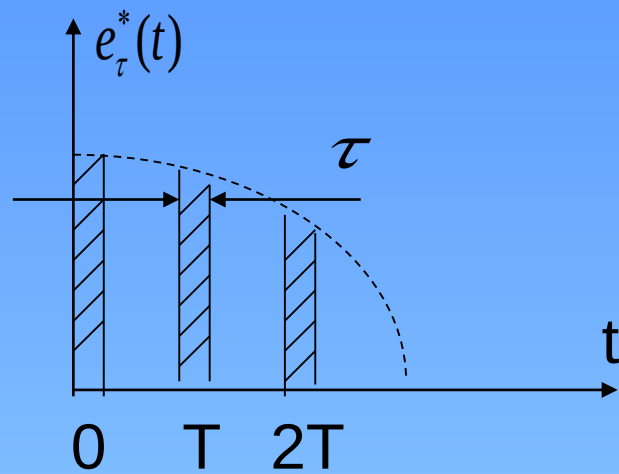
4、Sampling Signal

$$e^*(t) = \sum_{k=-\infty}^{\infty} e(t) \delta(t - kT)$$

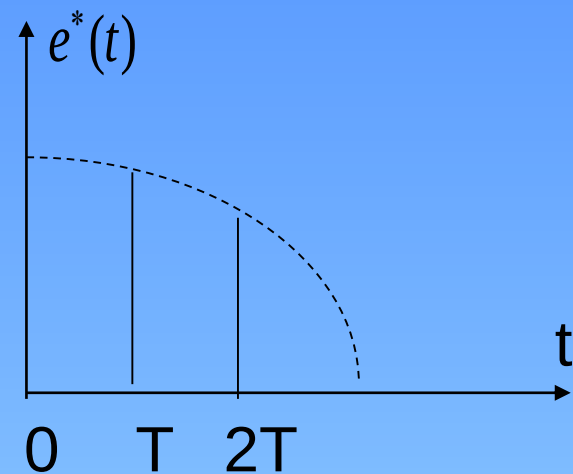
$$e^*(t) = \sum_{k=0}^{\infty} e(kT) \delta(t - kT)$$

real sampler





$\tau \rightarrow 0$



So the sampling operation can be expressed as

$$e^*(t) = \sum_{k=0}^{+\infty} e(kT) \cdot \delta(t - kT)$$

or

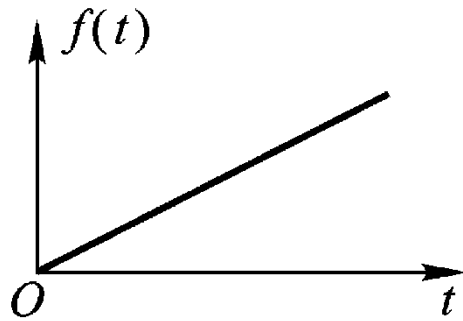
$$e^*(t) = e(t) \cdot \sum_{k=0}^{\infty} \delta(t - kT)$$

or

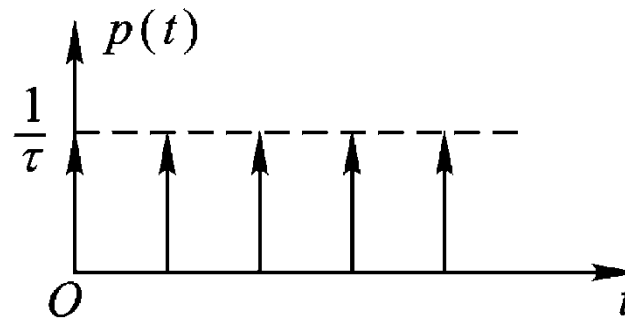
$$e^*(t) = e(t) \cdot \delta_T(t)$$

where

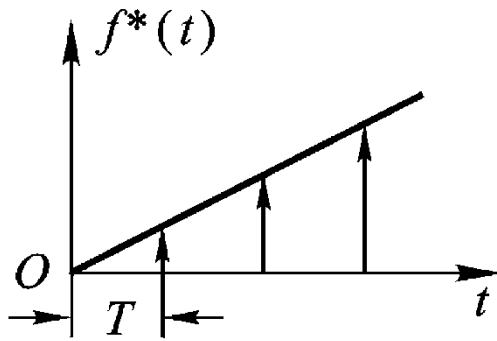
$$\delta_T(t) = \sum_{k=0}^{\infty} \delta(t - kT)$$



(a)



(b)



(c)



(d)

Fig 7—3 Sampling Process

$$f_{\tau}^*(t) = p(t) \cdot f(t)$$

Note:

$$\begin{aligned} e^*(t) &= e(0)\delta(t) + e(T)\delta(t-T) + \dots \\ &\neq e(0) + e(T) + \dots \end{aligned}$$

Laplace Transformation

位移定理:

- a. 实域中的位移定理，若原函数在时间上延迟 τ ，则其象函数应乘以 $e^{-\tau \cdot s}$

$$L[f(t - \tau)] = e^{-\tau \cdot s} F(s)$$

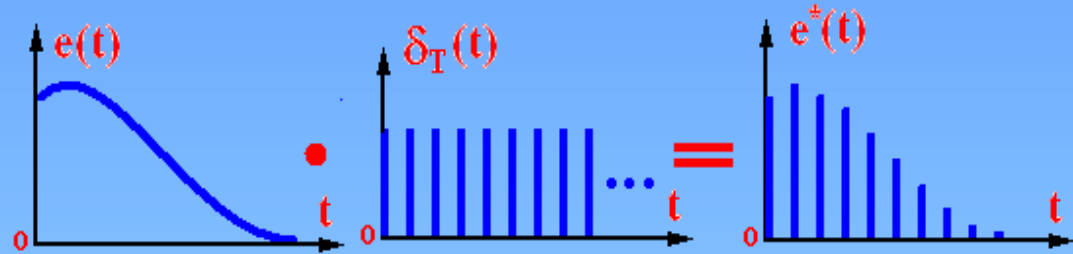
- b. 复域中的位移定理，象函数的自变量延迟 a ，原函数应乘以 e^{at} ，即

$$L[e^{at} f(t)] = F(s - a)$$

Ideal sampling sequence

$$\delta_T(t) = \sum_{n=0}^{\infty} \delta(t - nT)$$

$$e^*(t) = e(t) \cdot \delta_T(t)$$



$$= e(t) \cdot \sum_{n=0}^{\infty} \delta(t - nT) = \sum_{n=0}^{\infty} e(nT) \cdot \delta(t - nT)$$

$$(2) \mathcal{L}: \quad E^*(s) = \mathcal{L}[e^*(t)]$$

$$= \mathcal{L}\left[\sum_{n=0}^{\infty} e(nT) \cdot \delta(t - nT)\right] = \sum_{n=0}^{\infty} e(nT) \cdot e^{-nTs}$$

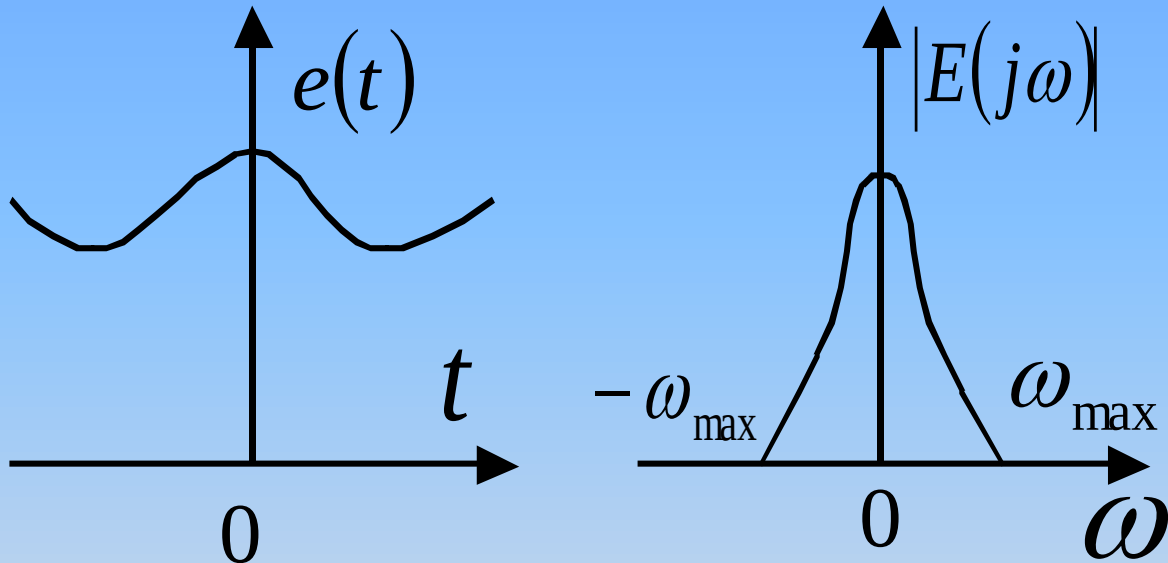
Example 7-1 $e(t) = 1(t)$ Obtain $E^*(s)$

Solution
$$E^*(s) = \sum_{n=0}^{\infty} 1 \cdot e^{-nTs}$$
$$= 1 + e^{-Ts} + e^{-2Ts} + \dots = \frac{1}{1 - e^{-Ts}} = \frac{e^{Ts}}{e^{Ts} - 1}$$

Example 7-2 $e(t) = e^{-at}$ Obtain $E^*(s)$

Solution
$$E^*(s) = \sum_{n=0}^{\infty} e^{-anT} \cdot e^{-nTs} = \sum_{n=0}^{\infty} e^{-(s+a)nT}$$
$$= \frac{1}{1 - e^{-(s+a)T}} = \frac{e^{Ts}}{e^{Ts} - e^{-aT}}$$

The continuous signal and its amplitude spectrum are

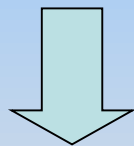


The Fourier-series expansion of $\delta_T(t)$:

$$\delta_T(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_s t}$$

$\omega_s = \frac{2\pi}{T}$ is the sampling freq.

$$c_k = \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} \delta_T(t) e^{-jk\omega_s t} dt \stackrel{\text{令 } t=0}{=} \frac{1}{T} \int_{0^-}^{0^+} \delta(t) dt = \frac{1}{T}$$



$$\delta_T(t) = \frac{1}{T} \sum_{k=-\infty}^{\infty} e^{jk\omega_s t}$$

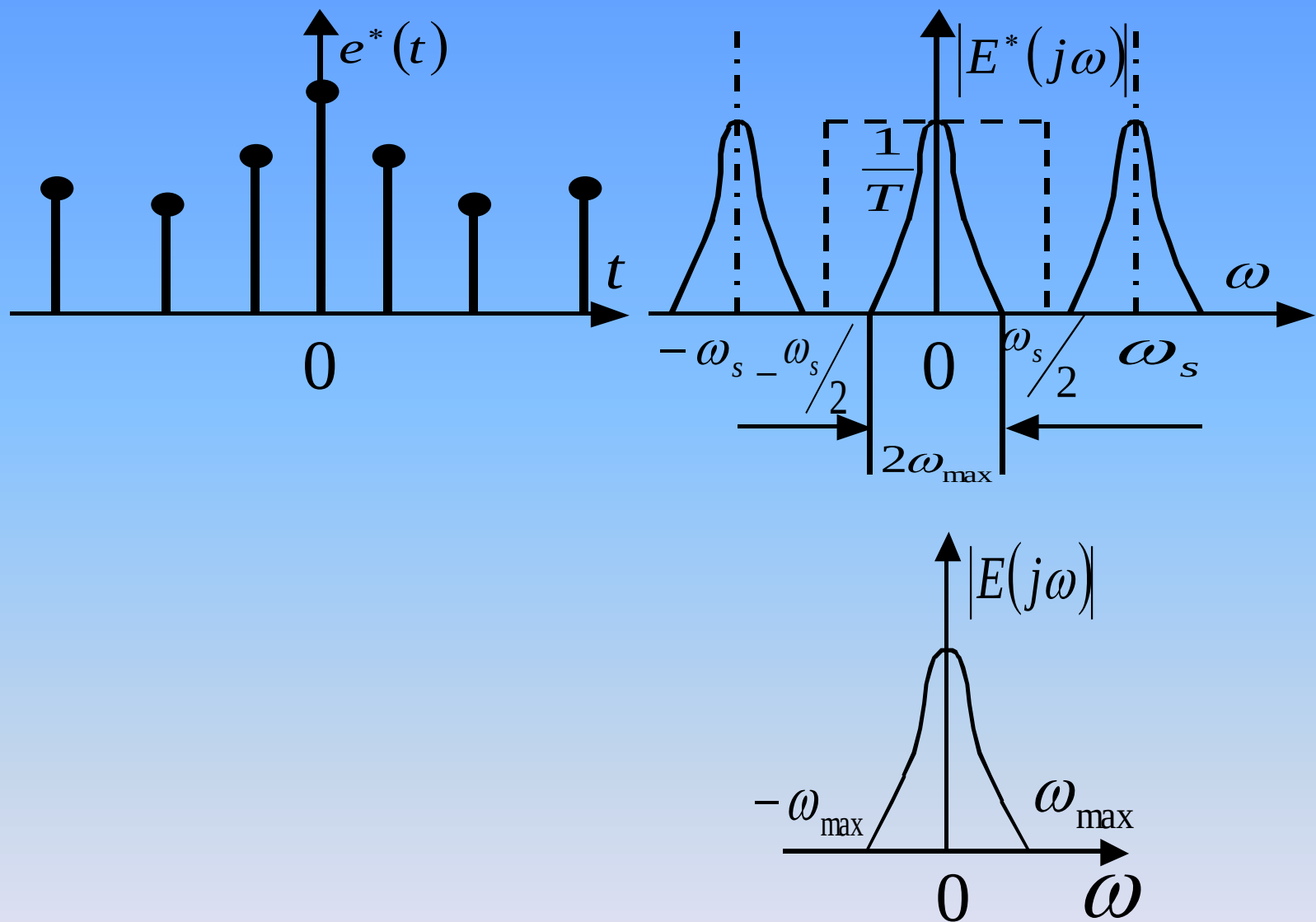
So the sampled signal is

$$e^*(t) = e(t) \cdot \delta_T(t) = \frac{1}{T} \cdot \sum_{k=-\infty}^{\infty} e(t) \cdot e^{jk\omega_s t}$$

which Laplace transform is

$$E^*(j\omega) = \frac{1}{T} \sum_{k=-\infty}^{+\infty} E[j(\omega + k\omega_s)]$$

where the operator s is replaced by $j\omega$



From the figure above, we can conclude that if $\omega_s > 2\omega_{\max}$

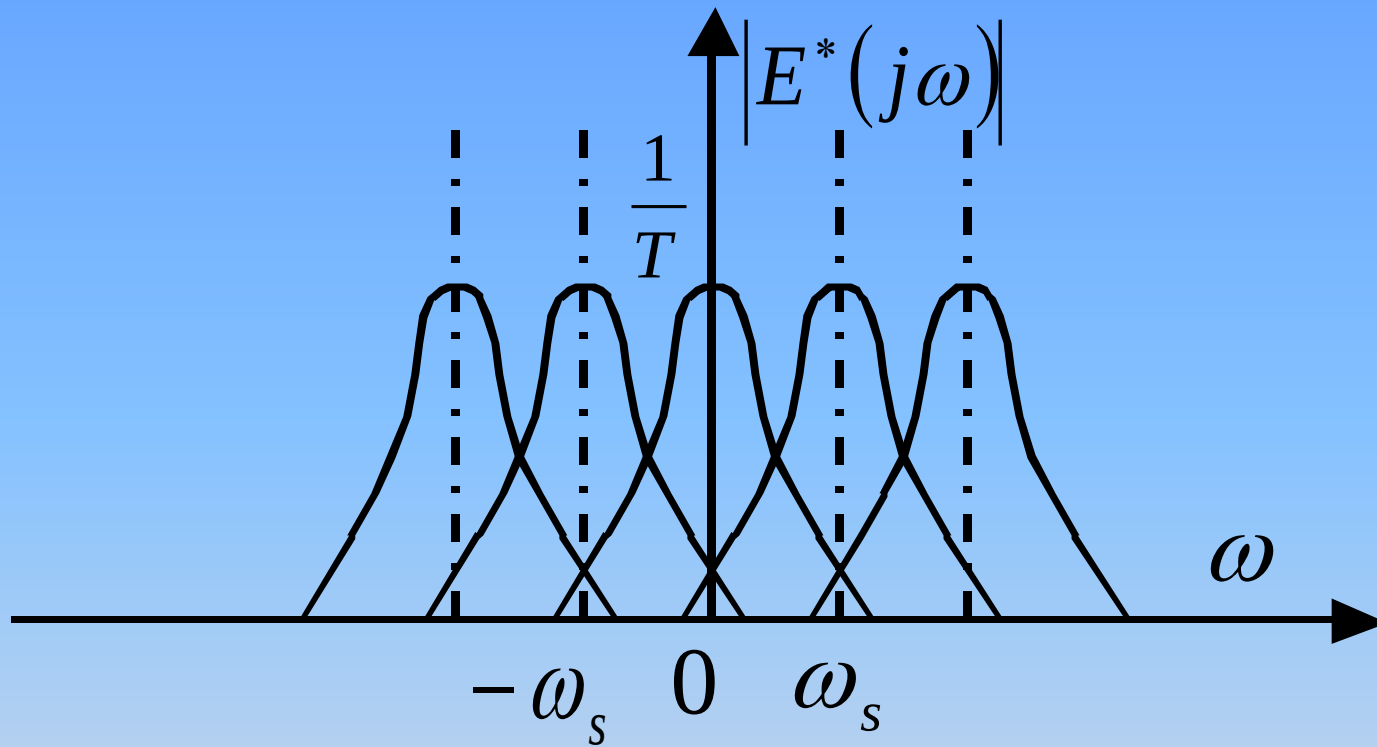
there are no overlap of each component, so the input signal can be recovered approximately. This is called sampling theorem or **Shannon's Theorem**

Shannon's Sampling Theorem:

Let $x(t)$ denote any continuous-time signal having a continuous Fourier transform

$$X(j\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t} dt$$

Let $x^*(t)$ denote the samples of $x(t)$ at uniform intervals of T seconds. Then $x(t)$ can be exactly reconstructed from its samples $x^*(t)$ if and only if $X(j\omega) = 0$ for all $|\omega| \geq \pi/T$

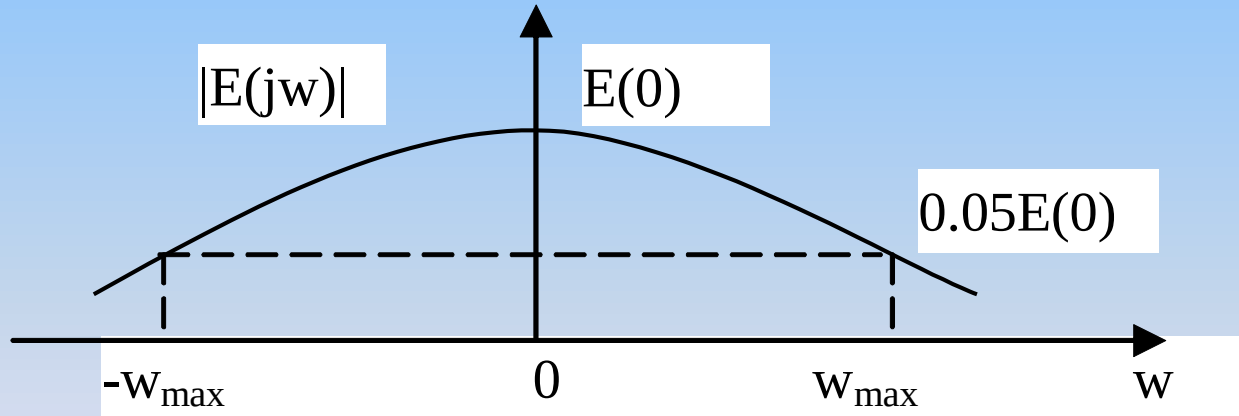


In the figure the input signal can't be recovered.

Problem: the maximum frequency $\omega_{\max}$ is *infinite* for a non-periodic signal!

Then, how could we select the sampling frequency ω_s for it?

Solution:



Example 7-3 Let $e(t)=e^{-t}$, select the sampling frequency according to Shannon's sampling theorem.

Solution.

The Laplace transform of $e(t)$: $E(s) = \frac{1}{s+1}$

The Fourier transform of $e(t)$ is: $E(j\omega) = \frac{1}{j\omega+1}$

The amplitude frequency characteristics:

$$|E(j\omega)| = \frac{1}{\sqrt{\omega^2 + 1}}$$

Assume the maximum frequency of $e(t)$ satisfies: $|E(j\omega_{\max})| = 0.05|E(0)|$

$$\frac{1}{\sqrt{\omega_{\max}^2 + 1}} = 0.05, \quad \omega_{\max} = 20 \text{ rad / s}$$

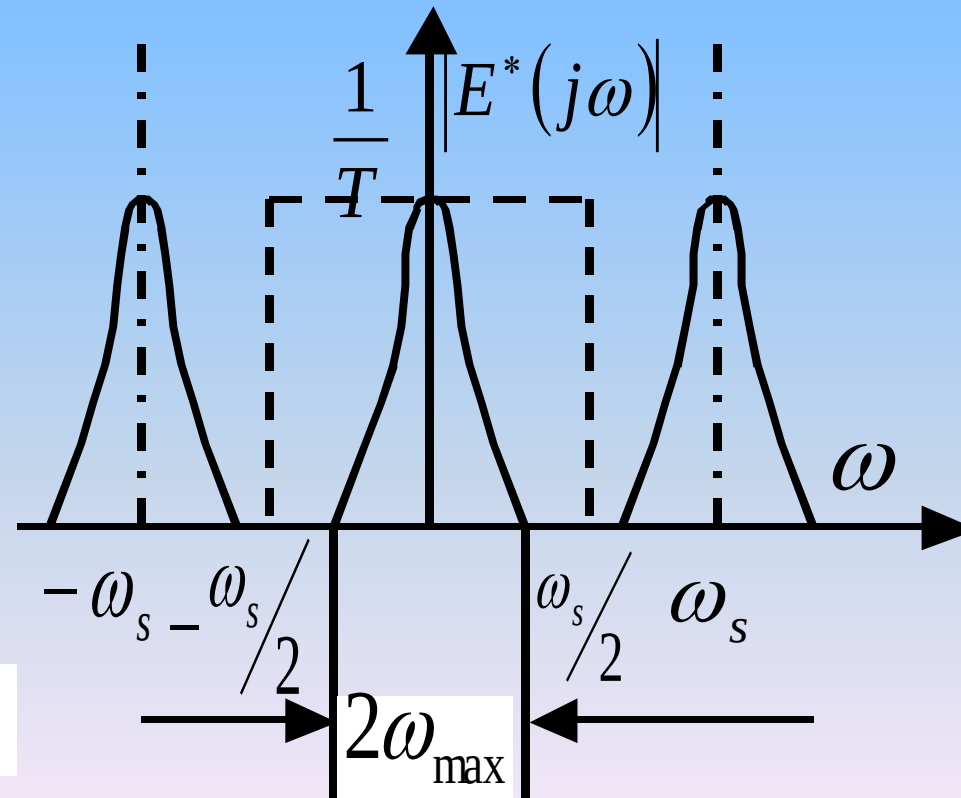
Then according to Shannon's theorem we have

$$\omega_s \geq 2\omega_{\max} = 40 \text{ rad / s}$$

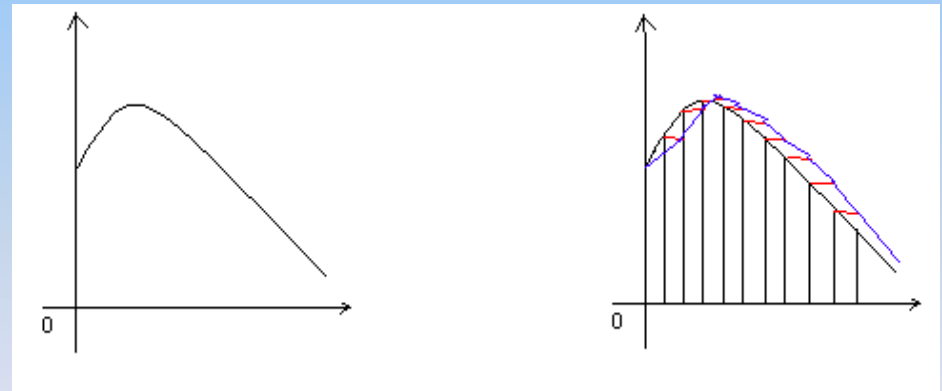
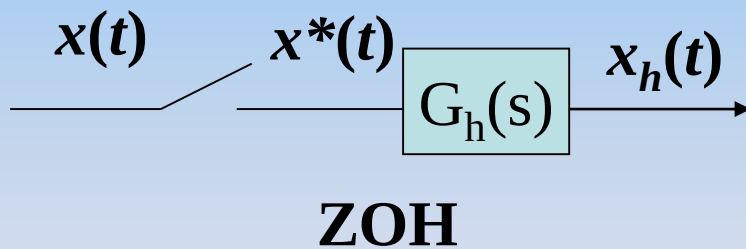
7.3 Signal Recovery and Zero-Order Hold

I. Signal recovery

The ideal filter is illustrated as the dotted line in the figure.



- The most common-used and simplest filter is zero-order hold filter.
- The zero-order hold (ZOH) is a mathematical model of the practical signal reconstruction done by a conventional digital-to-analog converter (DAC). That is, it describes the effect of converting a discrete-time signal to a continuous-time signal by *holding each sample value for one sample interval*.



Original signal $x(t)$

Recovered signal $x_h(t)$

The original signal $x(t)$ and recovered signal $x_h(t)$ satisfy :

$$x_h(t) = \sum_{k=0}^{\infty} x(kT)(1(t - kT) - 1(t - kT - T))$$

Apply Laplace transform on both sides of the equation, we have

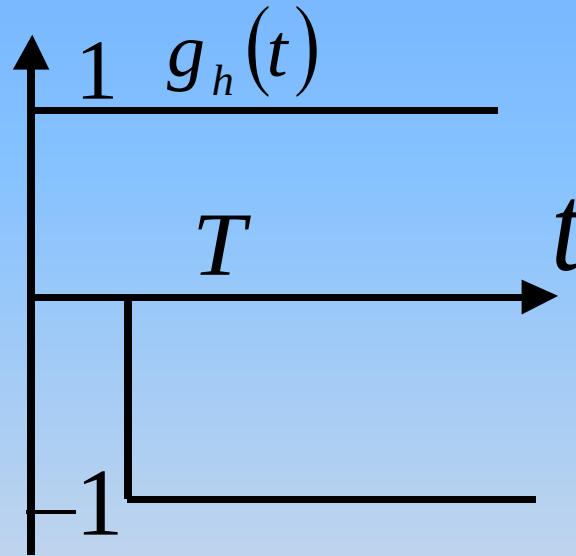
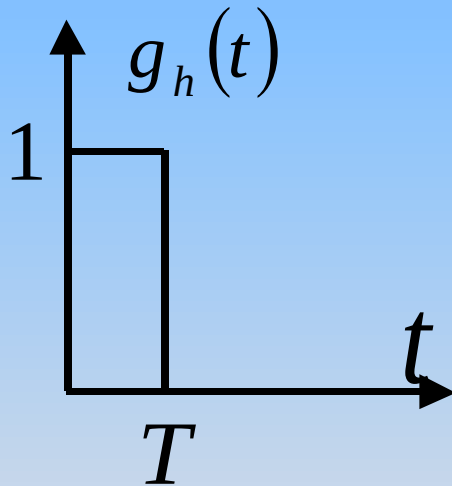
$$x_h(s) = \underbrace{\sum_{k=0}^{\infty} x(kT)e^{-kTs}}_{x^*(s)} \left[\frac{1}{s} - \frac{1}{s} e^{-Ts} \right]$$

Then the ZOH equivalent transition function is

$$\frac{x_h(s)}{x^*(s)} = \frac{1 - e^{-Ts}}{s} = G_h(s)$$

Corresponding time-domain function:

$$g_h(t) = 1(t) - 1(t - T)$$



Analysis of the frequency characteristics of ZOH filter :

$$G_h(j\omega) = \frac{1 - e^{-j\omega T}}{j\omega} = \frac{e^{-\frac{1}{2}j\omega T} (e^{j\frac{1}{2}\omega T} - e^{-j\frac{1}{2}\omega T})}{j\omega} = \frac{2e^{-\frac{1}{2}j\omega T} \left(\frac{e^{j\frac{1}{2}\omega T} - e^{-j\frac{1}{2}\omega T}}{2j} \right)}{\omega}$$

Considering we have, $\sin x = \frac{e^{jx} - e^{-jx}}{2j}$ therefore

$$G_h(j\omega) = \frac{2e^{-\frac{1}{2}j\omega T} \sin(\frac{1}{2}\omega T)}{\omega} \Rightarrow G_h(j\omega) = T \frac{\sin(\frac{\omega T}{2})}{\frac{\omega T}{2}} e^{-\frac{1}{2}j\omega T}$$

$$G_h(j\omega) = \frac{1 - e^{-j\omega T}}{j\omega} = T \cdot \frac{\sin(\omega T/2)}{\omega T/2} \cdot e^{-\frac{j\omega T}{2}}$$

$$\therefore T = \frac{2\pi}{\omega_s}$$

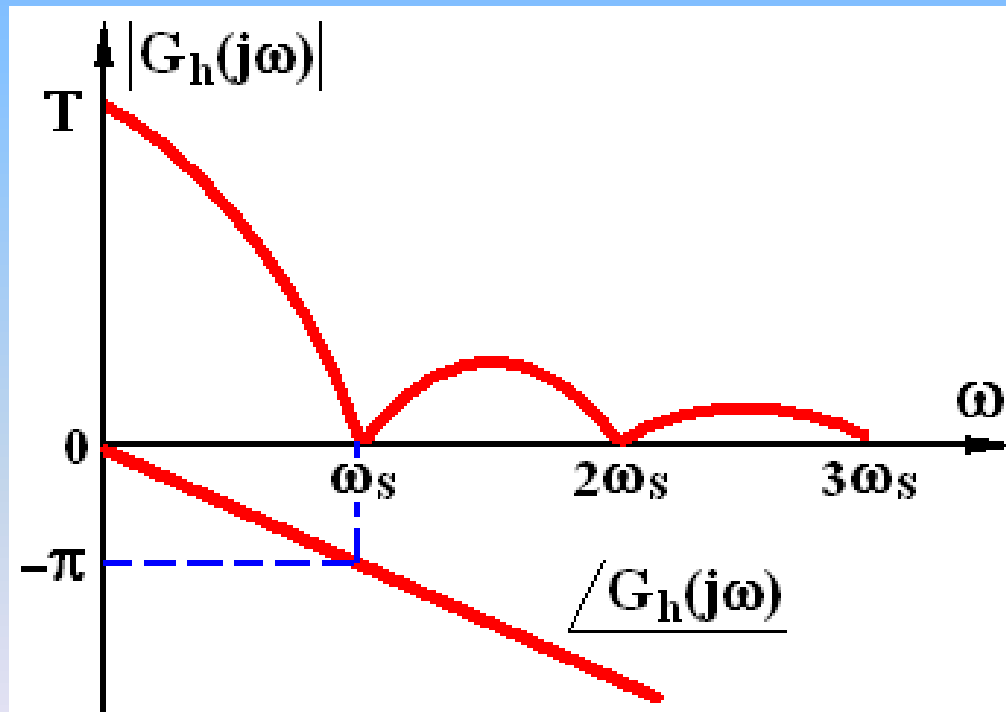
let $S_a(x) = \frac{\sin x}{x}$

then we have

$$G_h(j\omega) = \frac{2\pi}{\omega_s} \cdot S_a(\pi\omega/\omega_s) \cdot e^{-j\frac{\pi\omega}{\omega_s}}$$

Amplitude: $|G_h(j\omega)| = \frac{2\pi}{\omega_s} \cdot |S_a(\pi\omega/\omega_s)|$

Phase angle: $\angle G_h(j\omega) = -\frac{\pi\omega}{\omega_s}$



Note:

- The ZOH filter is not an ideal low-pass filter. ***Ripple error*** may occur after the filtering.
- There is a ***delayed phase angle*** when a signal is filtered by a ZOH.